

ASSIGNMENT #1

Course: CHE 341. CHEMICAL PROCESS CONTROL
Term: Spring 2020
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Your name: _____
Due Date: Wednesday January 22 by 5:00pm in my office

Number of points.....100
 Your score.....

TABLE #1

Symbol	Definition, Units
T_R	Reactor Temperature, C
V_R	Reactor volume, lit
C_{AA}	Acetic acid molarity, kmol/m ³
C_{ETOH}	Ethanol molarity, kmol/m ³
C_{EAC}	Ethyl acetate molarity, kmol/m ³
C_{WATER}	Water molarity, kmol/m ³
k_{fo}	Pre-exponential factor, m ³ /(kmol·s)
k_{ro}	Pre-exponential factor, m ³ /(kmol·s)
E_{fo}	Activation energy, kJ/mol
E_{ro}	Activation energy, kJ/mol
Symbol	Data for Problem 1B
k_{fo}	$1.9 \times 10^8 \text{ m}^3/(\text{kmol} \cdot \text{s})$
k_{ro}	$5.0 \times 10^7 \text{ m}^3/(\text{kmol} \cdot \text{s})$
E_{fo}	59.50 kJ/mol
E_{ro}	59.50 kJ/mol

Note: All problems are worth 100 points. Your final score will be the average value

Problem 1A: Review of Steady-State Mass and Energy Macroscopic Balances

The esterification of acetic acid with ethanol takes place in a water cooled Continuous Stirred Tank Reactor (CSTR) under atmospheric conditions and at a temperature of $\{T_R = 70^\circ C\}$. The reactor volume is $\{V_R = 500 \text{ lit}\}$ and the reaction mechanism and kinetic constants are given in the table below:

Reaction	$CH_3COOH + C_2H_5OH \xrightleftharpoons[k_r]{k_f} CH_3COOC_2H_5 + H_2O$
Reaction Rate	$\mathfrak{R}_1 = k_f C_{AA} C_{ETOH} - k_r C_{EAC} C_{WATER}$
Forward Rate Constant	$k_f = k_{f0} \exp\left(-\frac{E_f}{R_g T_R}\right)$
Reverse Rate Constant	$k_r = k_{r0} \exp\left(-\frac{E_r}{R_g T_R}\right)$

The conditions of the feed to the reactor are given in the table below:

Feed	Data
Temperature	$25^\circ C$
Pressure	1 atm
Acetic Acid Flow	193 kmol / hr
Ethanol Flow	187 kmol / hr
Ethyl Acetate Flow	0
Water Flow	10 kmol / hr

With these conditions pilot plant data indicate a conversion of the acetic acid equal to approximately 64%. The physical properties of the reaction mixture are described by the NRTL method for the liquid phase and by the ideal gas law for the vapor phase. **Prepare an ASPEN Plus simulation and report the following:**

- The heat and weight balance (Copy and Paste from the ASPN simulator)
- The density of the feed and product streams
- The heat capacity of the feed and product streams
- The heat of reaction
- The equilibrium constant. Verify whether the reaction has reached equilibrium or not
- The water flow rate required to cool the reactor
- The overall heat transfer coefficient between the reaction mixture and the cooling water

Show all required calculations in detail and report your results in a table format.

Problem 1B: Review of Unsteady-State Mass and Energy Macroscopic Balances.

Derive the equations of the evolution (as a function of time) of the species molar concentrations and the temperature of the reaction mixture. Show your derivations very clearly and carefully state all the assumptions that you make. Create a table of all the symbols that you use along with their numerical values of the various constants using data obtained from the steady-state ASPEN Plus simulation. Assume that the temperature of the water in the cooling jacket is maintained constant. The cooling water is supplied from a nearby cooling tower and returned back to the tower.

Implement your model in your favorite package (Matlab, or AthenaVisual Studio or Excel) and run the simulation for about one hour to obtain the steady state conditions of the reactor. Initially the reactor is filled with an equimolar mixture of acetic acid and water and heated up to about $\{T_o = 50^\circ\text{C}\}$. Check your steady-state values with those of ASPEN Plus to make sure that your simulation is correct. Your results should not be very different than the ones obtained by ASPEN. If you see large differences then it means that either your ASPEN Plus or your dynamic simulation is wrong. When everything is OK simulate the following scenario:

Because of an upstream disturbance in the water pumps twenty minutes after the reactor reached steady-state, the temperature of the cooling water in the reactor jacket dropped down by about $\{\Delta T = 5^\circ\text{C}\}$ for about twelve minutes before it was restored to its original value. **Perform a dynamic simulation and plot the reactor temperature profile as a function of time.**